

Customer Shopping Behaviour

❖ Data Analysis using python

1.Project Overview : This project analyzes customer shopping behavior by examining key demographic factors such as age group and gender. It identifies purchasing trends to understand which segments contribute most to sales. Seasonal patterns are evaluated to determine the highest purchasing periods throughout the year. The analysis helps uncover insights into customer preferences and buying motivations. These findings will support data-driven decisions for targeted marketing and improved sales strategies.

2.Dataset Summary : Dataset consists 3900 rows and 18 columns

Key feature 👍

Customer Details : Age, Gender, Location, Subscription

Purchase Details : Item purchased, Category, Purchase amount, Size, Seasons, Color

Shopping Behaviour : Discount Applied, Promo code used, Previous purchase, frequency purchase, Review Rating, Shipping type

–37 Value missing from Review Rating before Cleaning–

3.Exploratory Data Analysis using python:

1.Data Loading : Imported the dataset using pandas

2.Checking : Checking the data that its perfectly loaded using `df.head()`

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

3.Summary : Checking the dataset summary and structure using [df.info\(\)](#)

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	object
17	Frequency of Purchases	3900 non-null	object

dtypes: float64(1), int64(4), object(13)

memory usage: 548.6+ KB

4.Statistics : Analyze the Statistics using df.describe()

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

5.Null Value : Checked Dataset for Null value is there any null value so i used and count `df.isnull().sum()`

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season           0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

As we know that there are 37 missing value in Review Rating Column

6.Filled Null Value : We filled the missing value using group by category with median

7. Modification : First we did Columns Text Capitalize first word is upper case and rest of are lower case **df.columns.str.capitalize()**
Second we replace space with “_” using **df.columns.str.replace(“”,’_’)**

8.Mapping : To convert a string-based frequency purchase column (like "Day", "Week", "Month", "Year") into numerical values for analysis or modeling, you can use a mapping function in Python (Pandas). Here's a clean, scalable way to do it:

Example :

1.Day : 1

2.Week : 7

3.Month : 30

4.Quarter : 90

5.Year : 365

	purchase_frequency_days	Frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly
10	14	Bi-Weekly
11	14	Fortnightly
12	14	Fortnightly
13	7	Weekly
14	7	Weekly
15	30	Monthly
16	14	Bi-Weekly

❖ Data Analysis using Sql : Data loaded from jupyter notebook

1.Total Revenue between Male and Female

	Gender text	revenue numeric
1	Female	75191
2	Male	157890

2.Find those customer that used discount but they but they spent more than average purchase amount

	Customer_id bigint	Purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91

3.Top 5 product highest average rating

	Item_purchased text	Average_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.Compare the average purchase amount between standard and express shipping

	Shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5.Compare Average Spend and Total revenue between Subscriber and non-subscriber

	Subscription_status text	total_customer bigint	average_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645
2	No	2847	59.87	170436

6. Top 5 products have highest percentage with discount applied

	Item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. Checking our customer interaction from previous purchase if customer first time purchase then 1 else if customer more than 2 purchased then its called returning.

	customer_segments text	total_customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

8.Top 3 most product purchase within each category

	item_rank bigint 🔒	Category text 🔒	Item_purchased text 🔒	total_orders bigint 🔒
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

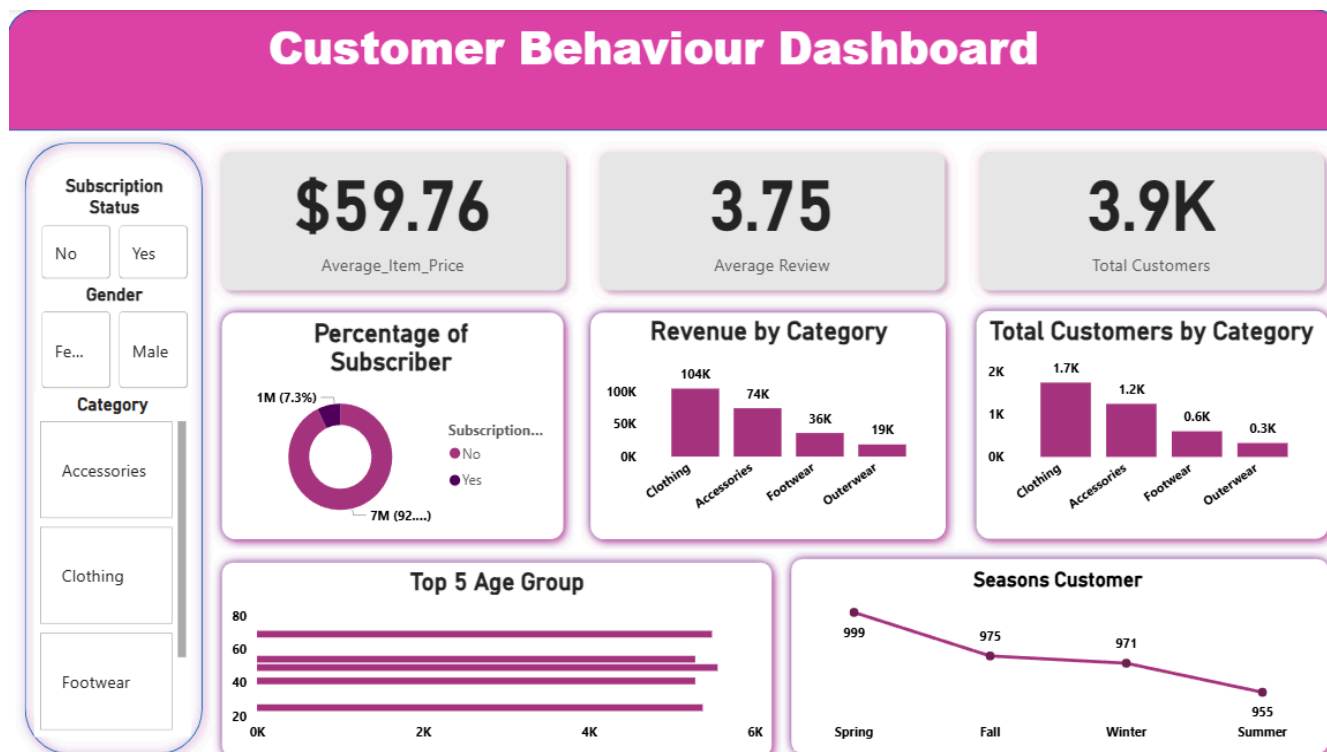
9.Customer who are repeat buyers more than 5 check from subscriber

	Subscription_status text 🔒	repeat_buyers bigint 🔒
1	No	2518
2	Yes	958

10.Distribution Revenue between age group

	Age bigint 	total_revenue numeric 
1	49	5552
2	69	5484
3	25	5372
4	41	5282
5	54	5282
6	57	5200
7	28	5104
8	19	4941
9	50	4930
10	31	4864
11	62	4809
12	32	4796
13	53	4752
14	37	4737

❖ Customer Behavior Dashboard



Key points:

- 1.Total Customer :** 3900
- 2.Average Review :** 3.75
- 3.Average Item price :** 59.76
- 4.Subscriber Status :** Around 7.3% who don't have Subscription and 93.7% Have subscription
- 5.Revenue of Category :** Clothing Has revenue generator and outerwear has lowest revenue
- 6.Total Customer by Category :** Again Clothing interact more customer
- 7.Top Age Group Buyers :** Aged 49 has Highest Distribution in Revenue
- 8.Seasons Customer :** Customer buying more items in Spring Seasons and summer seasons is not customer more customer is average

